

# DUAL-CORE OPTICAL MOTION TESTBED

## FIRMWARE, HOST SOFTWARE & RESULTS

STM32H747 dual-core motion control, live telemetry, automated test execution, and measured characterization

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 5.0 | 5 AUGUST 2026

|                                                                                                                                                                                                                                                                                                                           |                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Prepared by                                                                                                                                                                                                                                                                                                               | Herder Elektronische Systemen                                                                         |
| Document scope                                                                                                                                                                                                                                                                                                            | Dual-core firmware, IPC, host GUI, telemetry integrity, safeguards, and measured motion performance   |
| Evidence basis                                                                                                                                                                                                                                                                                                            | MotionIPC source, target firmware, GUI recordings, plots, logs, test scripts, and engineering reviews |
| Distribution                                                                                                                                                                                                                                                                                                              | Customer-facing; controlled source excerpts and engineering evidence                                  |
| <div>DOCUMENT PURPOSE</div> <div>Defines the controlled functional, performance, interface, safety, verification, and acceptance requirements for the dual-core target firmware and host-side test system. Mechanical construction and PCB manufacturing details are referenced to the companion platform document.</div> |                                                                                                       |

# Functional Requirements Document Framework

Why this is an FRD

This report defines what the subsystem shall do, the interfaces and constraints under which it shall operate, and the objective evidence used to determine whether each requirement is satisfied. Photographs, schematics, plots, and source-code observations support the requirements; they do not replace them.

## Purpose, Scope, and System Boundary

This FRD governs the dual-core target firmware, interprocessor communication, host application, telemetry, automated test execution, data retention, safeguards, and measured motion-performance reporting. The mechanical structure and interface PCB are external interfaces documented in the companion platform FRD.

| Boundary element | Included in this FRD                                                                 | Excluded / referenced                                              |
|------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Target firmware  | M7/M4 workload allocation, motion sequencing, timing, IPC, heartbeat, fault state    | Mechanical fabrication and PCB manufacturing details               |
| Host software    | Connection control, commands, live plots, test automation, CSV and metadata          | General-purpose IT infrastructure and cloud services               |
| Verification     | Low-speed tracking, constant-speed, step response, sine-response, integrity counters | Formal environmental, EMC, safety, or optical-system certification |

## Users and Operational Scenarios

| Actor              | Primary interaction                                          | Expected outcome                                                           |
|--------------------|--------------------------------------------------------------|----------------------------------------------------------------------------|
| Operator           | Connect, configure, command, monitor, stop, and save a test  | Controlled motion with visible state and retained evidence                 |
| Firmware developer | Build, deploy, inspect IPC and timing behavior               | Traceable target revision and diagnosable inter-core operation             |
| Test engineer      | Select scripted characterization and review outputs          | Repeatable datasets suitable for comparison and analysis                   |
| Reviewer/customer  | Read requirements, figures, limits, and verification results | Clear distinction between demonstrated capability and open release actions |

# Functional Requirements

| ID        | The system shall...                                                                   | Rationale / explanation                                                                    | Verification                                         |
|-----------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------|
| FW-FR-001 | separate low-rate control/status messaging from high-rate motion-record transfer.     | Control messages must remain intelligible while motion data moves continuously.            | Architecture review; IPC source inspection; live run |
| FW-FR-002 | publish motion records with sequence and integrity information through shared memory. | Sequence, overflow, and heartbeat fields make loss or stalling observable.                 | Memory-map review; counter observation               |
| FW-FR-003 | accept operator motion commands and expose commanded and measured state               | The operator needs direct confirmation that target behavior follows the requested profile. | GUI demonstration; captured telemetry plots          |

| ID        | The system shall...                                                                            | Rationale / explanation                                                                               | Verification                              |
|-----------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------|
|           | in the host GUI.                                                                               |                                                                                                       |                                           |
| FW-FR-004 | execute repeatable low-speed, constant-speed, step, and sine-response tests.                   | Scripted tests reduce operator variation and preserve comparable datasets.                            | Test selection and completed result files |
| FW-FR-005 | record CSV telemetry and run metadata sufficient to reproduce the analysis context.            | Plots must remain traceable to commands, timestamps, configuration, and source data.                  | CSV/header inspection; metadata review    |
| FW-FR-006 | limit pulse generation and bound catch-up behavior.                                            | A delayed software loop must not produce an uncontrolled pulse burst.                                 | Source inspection; stress test            |
| FW-FR-007 | detect missing host presence and expose target heartbeat, fault, sequence-gap, and drop state. | Communication silence must be distinguishable from normal idle operation.                             | Fault-injection and disconnect test       |
| FW-FR-008 | preserve a safe stop path independent of normal scripted completion.                           | The operator must be able to terminate motion when a test or communication path behaves unexpectedly. | Stop-command and deadman test             |

## Performance and Quality Requirements

| ID         | Requirement                                                                                                  | Acceptance basis                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| FW-NFR-001 | Motion pulse generation shall not exceed 25,000 pulses/s in the reviewed configuration.                      | Source constant and measured output remain at or below the cap.                                       |
| FW-NFR-002 | STEP high time shall be at least 4 microseconds.                                                             | Logic trace or source review confirms the minimum.                                                    |
| FW-NFR-003 | Host-presence timeout shall stop or inhibit continued commanded operation after 2 seconds without heartbeat. | Disconnect test produces the defined safe response.                                                   |
| FW-NFR-004 | Telemetry-rate selections shall be explicit and the sustainable serial rate shall not be overstated.         | 10/20/50/100 Hz options are visible; report distinguishes configured from demonstrated throughput.    |
| FW-NFR-005 | Shared-memory records shall maintain cache-safe alignment and bounded service behavior.                      | 64-byte alignment and 128-record service bound confirmed in source.                                   |
| FW-NFR-006 | Reported performance shall identify test conditions, limits, and unsupported claims.                         | Result sections state the demonstrated range and avoid unsupported bandwidth or certification claims. |

## External Interface Requirements

| Interface                  | Required behavior                                                              | Failure / boundary condition                                                |
|----------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| STEP/DIR output            | Generate direction state and bounded step pulses for the motor driver.         | Rate cap, minimum high time, and safe stop remain enforced.                 |
| Differential encoder input | Receive conditioned encoder position/count information from the interface PCB. | Missing or implausible motion shall be visible in measured-state telemetry. |
| Host serial link           | Transport commands, status, telemetry, and heartbeat.                          | Loss of heartbeat invokes the defined timeout behavior.                     |

| Interface     | Required behavior                                                  | Failure / boundary condition                                          |
|---------------|--------------------------------------------------------------------|-----------------------------------------------------------------------|
| OpenAMP/RPMsg | Carry low-rate command and status messages between cores.          | Endpoint or message failure is reported rather than silently ignored. |
| AXI-SRAM ring | Carry aligned high-rate records with producer/consumer accounting. | Overflow, dropped records, and sequence gaps remain observable.       |

1. System Role and Delivered Capability

The system combines a dual-core STM32H747 target with a host application that commands motion, displays live encoder and speed telemetry, executes repeatable tests, and writes analysis-ready records. The public evidence package separates control-plane messaging from high-rate motion-data movement and preserves the distinction between demonstrated behavior and recommended release actions.

- Cortex-M7 and Cortex-M4 workloads connected through OpenAMP/RPMsg and a custom AXI-SRAM shared ring.
- Live host console for connection, scripted test selection, telemetry plots, CSV logging, and metadata capture.
- Measured low-speed tracking, constant-speed error, step response, and closed-loop frequency-response evidence.
- Fault, heartbeat, sequence-gap, dropped-record, and ring-status observability.

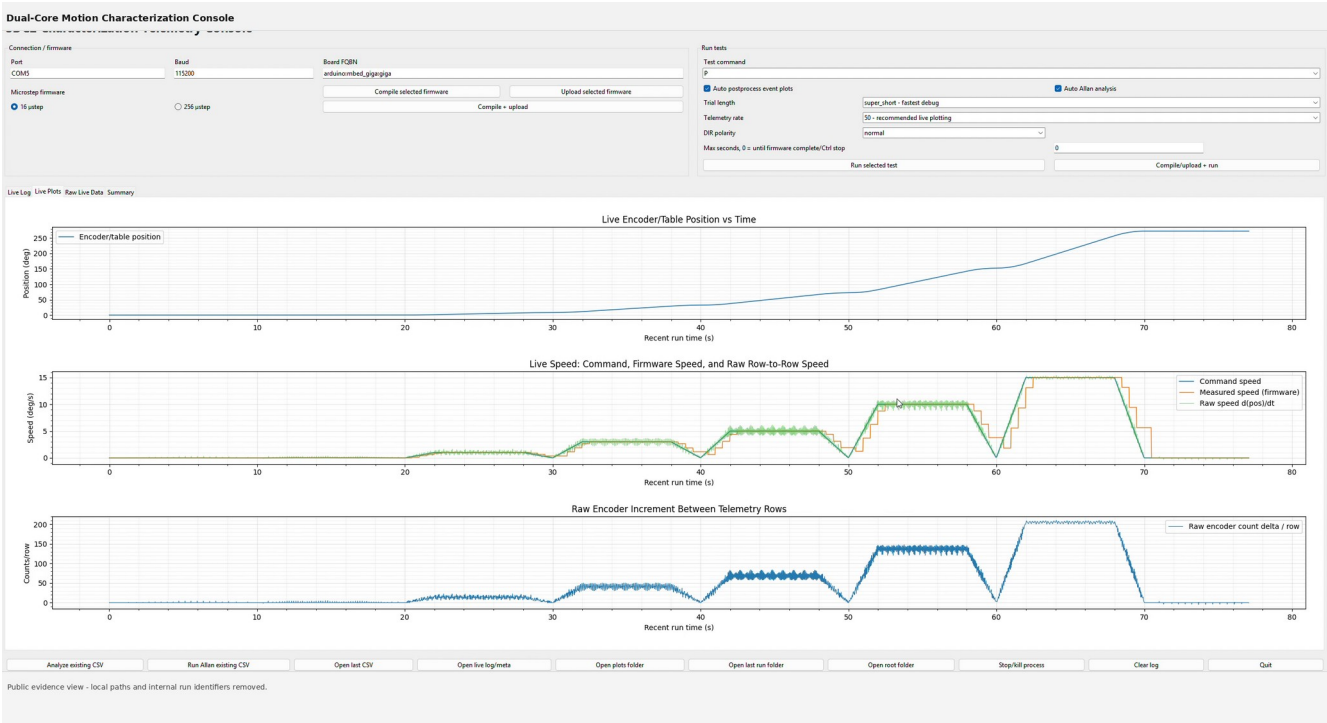


Figure 1. Live host telemetry console during an automated characterization sequence.

## 2. Dual-Core Firmware Architecture

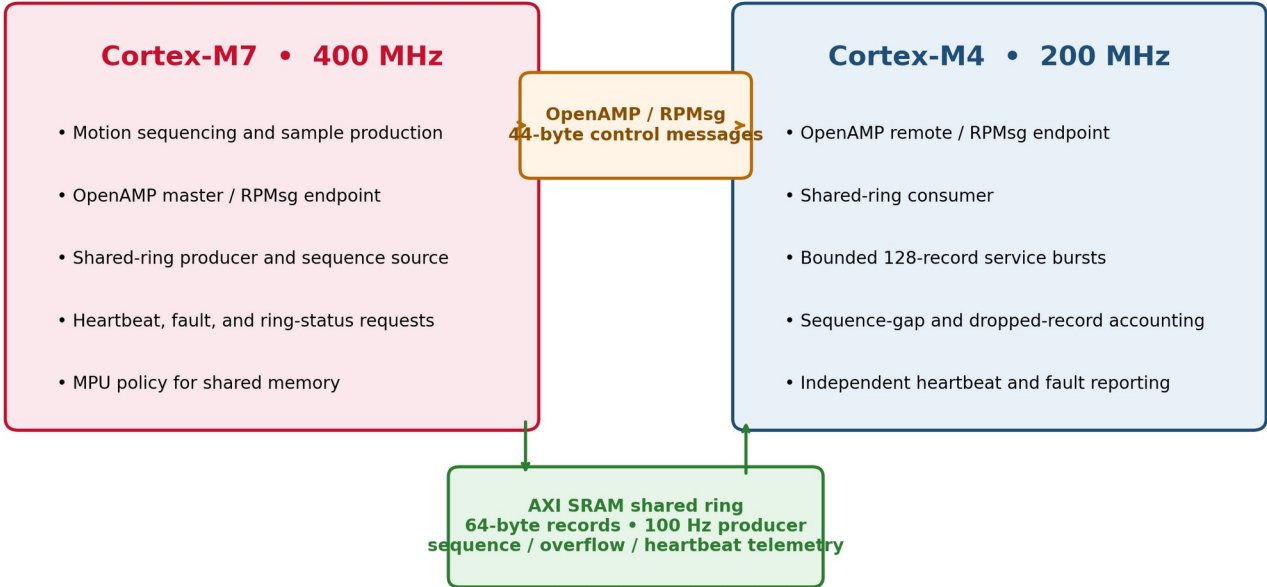


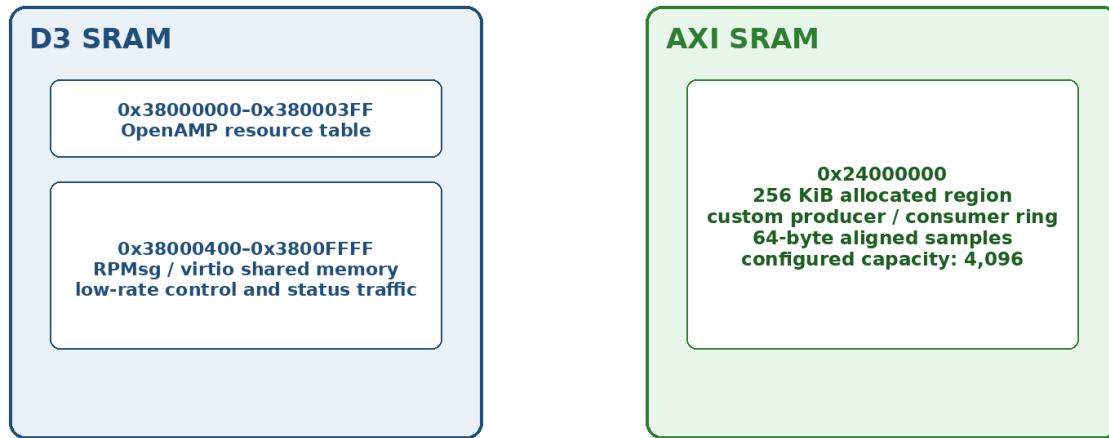
Figure 2. Dual-core architecture with physically separated control-message and shared-ring paths.

| Interface       | Verified role                            | Representative parameter                         |
|-----------------|------------------------------------------|--------------------------------------------------|
| OpenAMP / RPMsg | Low-rate command and status coordination | 44-byte application message                      |
| AXI-SRAM ring   | High-rate aligned motion records         | 64-byte records; 4,096 nominal entries           |
| M7 producer     | Motion sequencing and sample publication | 100 Hz sample production in the reviewed package |
| M4 consumer     | Bounded service and accounting           | Up to 128 records per service burst              |

### 3. Shared Memory, Timing, and Integrity

#### MotionIPC Shared-Memory Map

Separate regions preserve ST OpenAMP resources while providing a high-throughput custom data path



**Engineering review finding: 4,096 × 64-byte records plus metadata exceed the 256 KiB allocation by ~64 by**

*Figure 3. MotionIPC shared-memory allocation and the engineering review finding on nominal ring capacity.*

- OpenAMP resources occupy D3 SRAM while the custom producer/consumer ring is allocated in AXI SRAM.
- Records are aligned to cache-line boundaries and include sequence, overflow, and heartbeat information.
- Engineering review identified that 4,096 records plus metadata exceed a 256 KiB allocation by approximately one record; release should use 4,095 entries or enlarge the reserved region.
- The target reports independent heartbeat and fault state so host-side silence is distinguishable from target-side failure.

4. Host Application and Test Workflow

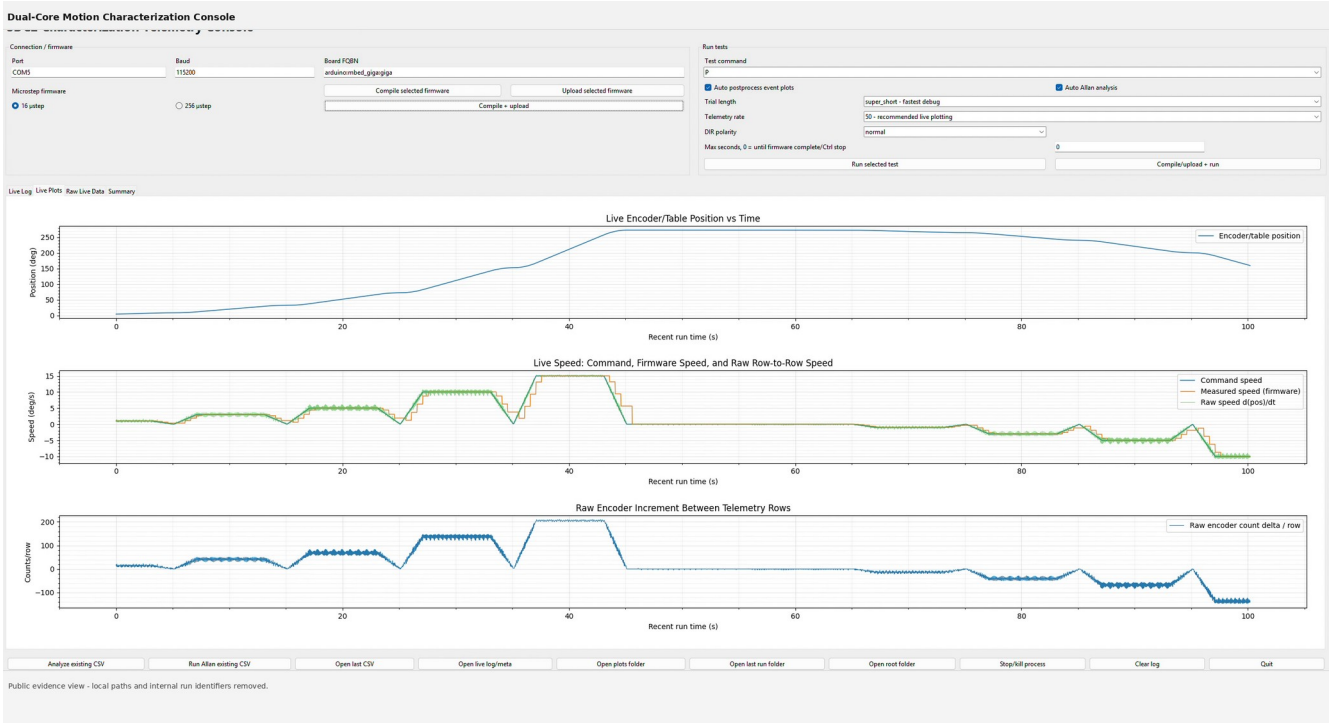


Figure 4. Host console view showing live position, commanded/measured speed, and raw encoder increments.

| Workflow stage | Operator action                     | Recorded output                         |
|----------------|-------------------------------------|-----------------------------------------|
| Connect        | Select serial port and baud rate    | Target identity and connection state    |
| Configure      | Select firmware and test definition | Run metadata and configuration          |
| Execute        | Start automated sequence            | Live plots and telemetry rows           |
| Review         | Open generated outputs              | CSV, figures, logs, and summary metrics |

## 5. Low-Speed Tracking Results

SDC2 ONLY | 256 MICROSTEP BASELINE

### Speed tracking summary: usable ultra-slow range, clear high-speed ceiling

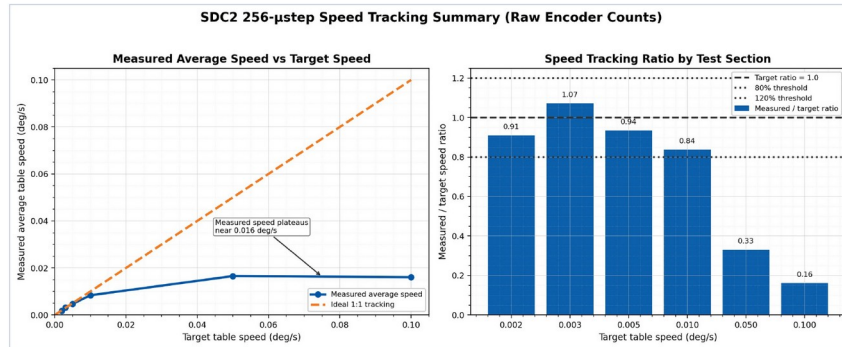


Figure 1. Clean summary view with legend moved off the data region. Average speed is computed from total raw encoder count delta over each section.

#### Main result

- 0.002-0.01 deg/s tracks in the correct range at the current 256 microstep setting.
- Best tracking occurs around 0.003-0.005 deg/s; 0.01 deg/s is usable but below target.
- 0.05 and 0.10 deg/s fail similarly and plateau near 0.016 deg/s actual speed.
- Interpretation: higher commanded speeds likely require lower microstepping and/or controller tuning.

#### Numerical baseline

- 0.002 -> 0.001818 deg/s, ratio 0.909
- 0.003 -> 0.003214 deg/s, ratio 1.071
- 0.005 -> 0.004676 deg/s, ratio 0.935
- 0.010 -> 0.008378 deg/s, ratio 0.838
- 0.050 -> 0.016560 deg/s, ratio 0.331
- 0.100 -> 0.016071 deg/s, ratio 0.161

SDC2 Encoder-Based Speed Tracking

Raw encoder-event analysis | no filtering or grouped averaging

1

Figure 5. Speed-tracking summary showing the usable ultra-slow range and high-speed ceiling.

| Commanded speed (deg/s) | Measured speed (deg/s) | Interpretation                           |
|-------------------------|------------------------|------------------------------------------|
| 0.002                   | 0.001818               | Trackable near the lower practical limit |
| 0.003                   | 0.003214               | Best agreement region                    |
| 0.005                   | 0.004676               | Strong low-speed tracking                |
| 0.010                   | 0.008378               | Usable with increasing bias              |
| 0.050-0.100             | ~0.016 plateau         | Configuration-limited ceiling            |



## 6. Constant-Speed and Step-Response Characterization

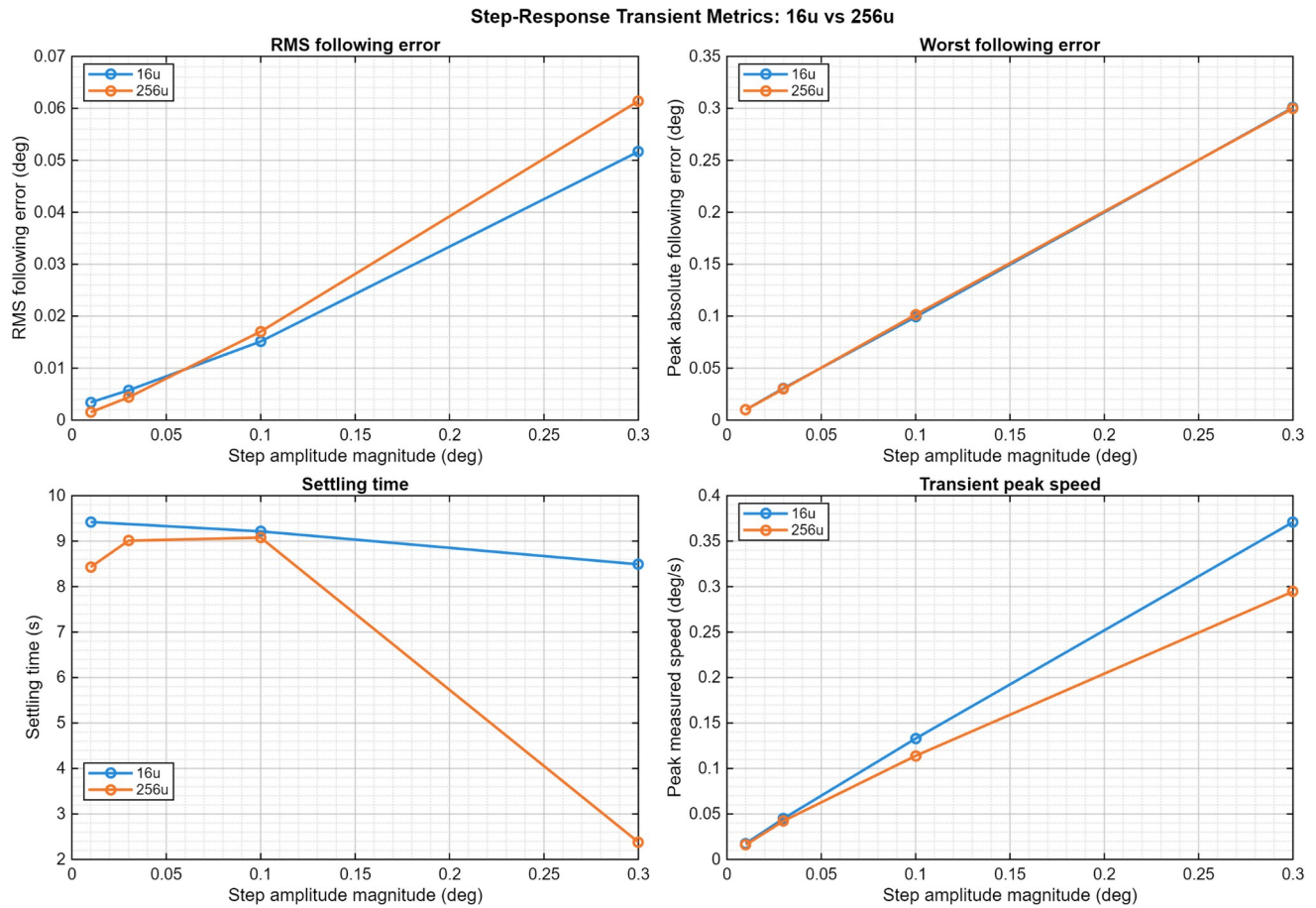


Figure 6. Extended constant-speed tracking and Allan-style stability summary.

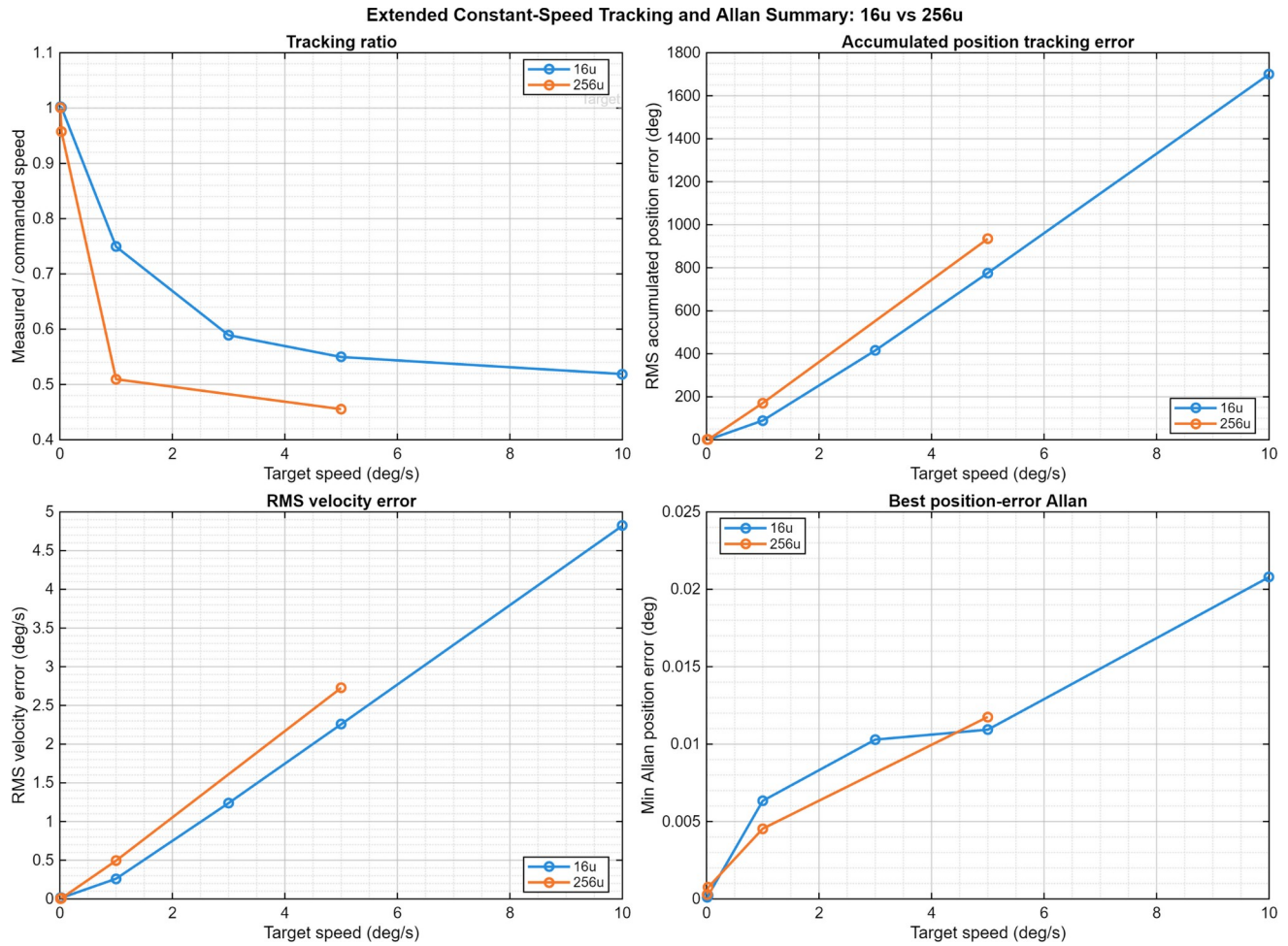


Figure 7. Step-response metrics for RMS following error, worst following error, gain, and position-area measures.

## 7. Closed-Loop Frequency Response

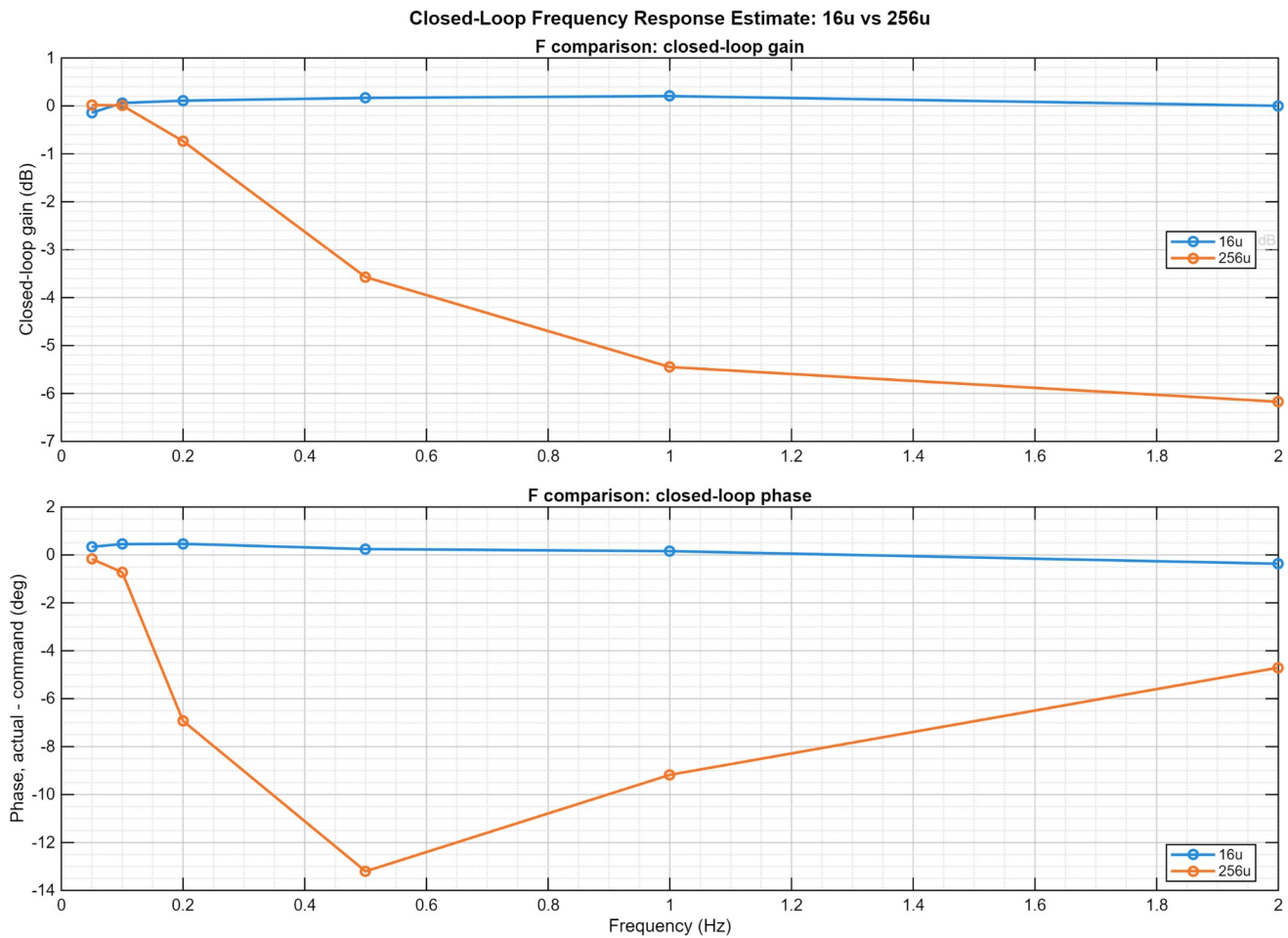


Figure 8. Closed-loop response estimate across the tested 0.33-2 Hz range.

The available evidence supports reporting response across the tested frequencies. It does not, by itself, establish a formal -3 dB bandwidth because the plotted data do not demonstrate a complete crossing with sufficient surrounding points.

## 8. Detailed Firmware and Host Design

The following parameters are retained as engineering design detail rather than reduced to marketing-level summaries. They define the implemented timing model, inter-core transport, motion conversion, serial limits, safety behavior, and measured operating envelope.

| Subsystem         | Implemented detail                                                                            | Engineering significance                                                               |
|-------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Processor split   | STM32H747 Cortex-M7 at 400 MHz and Cortex-M4 at 200 MHz                                       | Separates motion-data production from remote consumption and host-facing service work. |
| Control messaging | OpenAMP/RPMsg; 44-byte application message with eight 32-bit payload words                    | Low-rate commands and status remain distinct from high-rate motion records.            |
| High-rate path    | AXI-SRAM producer/consumer ring at 0x24000000; 64-byte records aligned to 32-byte cache lines | Supports deterministic bulk transfer with sequence and overflow observability.         |
| Resource table    | OpenAMP shared resources in D3 SRAM beginning at 0x38000400                                   | Keeps RPMsg resources separate from the custom motion-data ring.                       |
| Service behavior  | M4 consumes bounded bursts of up to 128 records per pass                                      | Prevents unbounded ring service from starving other work.                              |
| Health reporting  | Independent heartbeats, fault state, sequence-gap, dropped-record, and ring-status counters   | Allows the host to distinguish silence, backlog, overflow, and target faults.          |

### 8.1 Motion Timing and Safety Parameters

| Parameter             | Value                                   | Notes                                                                                       |
|-----------------------|-----------------------------------------|---------------------------------------------------------------------------------------------|
| Serial link           | 115,200 baud                            | Full 24-field ASCII rows make 100 Hz a configured option, not a guaranteed sustained rate.  |
| Telemetry selections  | 10, 20, 50, or 100 Hz; clamped 1-100 Hz | Practical sustained rate depends on row length and host/target overhead.                    |
| STEP pulse high time  | 4 microseconds                          | Explicit pulse-width control for the stepper driver interface.                              |
| Maximum step rate     | 25,000 pulses/s                         | Hard safety ceiling; equivalent to approximately 7.13 deg/s with the documented drivetrain. |
| Backlog limit         | 25 pulses emitted per loop              | Bounds catch-up behavior after temporary host or loop delay.                                |
| Overrun warning       | 25 ms                                   | Flags excessive loop latency.                                                               |
| Host heartbeat        | 500 ms                                  | Periodic host liveness message.                                                             |
| Target deadman        | 2 s                                     | Stops or faults commanded operation when host supervision is lost.                          |
| Speed estimate window | 500 ms                                  | Reduces encoder quantization noise in ultra-low-speed operation.                            |

### 8.2 Drivetrain and Encoder Conversion

| Quantity           | Calculation / value                 | Interpretation                                   |
|--------------------|-------------------------------------|--------------------------------------------------|
| Motor steps        | 200 full steps/rev x 256 microsteps | 51,200 pulses per motor revolution.              |
| Gear ratio         | 24.65:1                             | Produces 1,262,080 pulses per table revolution.  |
| Command resolution | 3,505.78 pulses/degree              | Approximately 0.000285 degree per command pulse. |

| Quantity            | Calculation / value                            | Interpretation                                                 |
|---------------------|------------------------------------------------|----------------------------------------------------------------|
| Encoder decoding    | 5,000 CPR, A-channel CHANGE x2                 | 10,000 motor-shaft counts/rev.                                 |
| Table encoder scale | 246,500 counts/rev                             | 684.72 counts/degree, approximately 0.001460 degree/count.     |
| Correction gain     | 20 pulse-Hz/degree                             | Closed-loop correction term applied to feedforward command.    |
| Correction limits   | 250 Hz absolute and 5% of feedforward relative | Prevents the correction term from dominating commanded motion. |

### 8.3 Measured Operating Results

| Commanded speed   | Measured speed                    | Observation                                                                            |
|-------------------|-----------------------------------|----------------------------------------------------------------------------------------|
| 0.002 deg/s       | 0.001818 deg/s                    | Lower edge of demonstrated ultra-slow tracking.                                        |
| 0.003 deg/s       | 0.003214 deg/s                    | Best tracking region begins.                                                           |
| 0.005 deg/s       | 0.004676 deg/s                    | Strong agreement in the preferred low-speed range.                                     |
| 0.010 deg/s       | 0.008378 deg/s                    | Tracking remains usable but error grows.                                               |
| 0.050-0.100 deg/s | approximately 0.016 deg/s plateau | Observed ceiling in the tested 256-microstep configuration.                            |
| Sine tests        | 0.33, 0.5, 1, and 2 Hz            | Supports response reporting over the tested range, not a formal -3 dB bandwidth claim. |

### 8.4 Host Workflow and Data Products

- The host application provides connection setup, command entry, scripted run selection, live position and speed plots, raw encoder-increment visibility, CSV logging, metadata capture, and post-run analysis launch points.
- Each run preserves command values, measured telemetry, timing, run identity, and analysis context so plots can be traced back to the source recording rather than treated as disconnected screenshots.
- Automated characterization covers low-speed sweeps, constant-speed tracking, step response, and closed-loop sine-response tests; safety and heartbeat state remain visible during those operations.

## 9. Safety, Reliability, and Release Controls

| Control          | Implemented behavior                            | Release consideration                          |
|------------------|-------------------------------------------------|------------------------------------------------|
| Host heartbeat   | Periodic host presence message                  | Target deadman should remain enabled           |
| Pulse-rate cap   | 25,000 pulses/s maximum                         | Verify against final drivetrain and driver     |
| Backlog bound    | Maximum 25 pulses emitted per loop              | Prevents unlimited catch-up bursts             |
| Telemetry limits | Selectable 10/20/50/100 Hz                      | Full ASCII rows may constrain sustainable rate |
| Fault accounting | Sequence gaps, drops, heartbeat, fault counters | Expose in final operator status panel          |

At 115,200 baud, full 24-field ASCII telemetry rows cannot be assumed to sustain 100 Hz under all conditions. The configured option should be treated separately from demonstrated end-to-end serial throughput; 50-60 Hz is a more realistic full-row operating range without a binary protocol or higher link rate.



## Requirements Verification and Traceability

The matrix below links each requirement to objective evidence and a current disposition. “Demonstrated” means evidence is present in this report or its cited source package; “Release action” means the requirement remains to be completed or formally witnessed before a controlled production release.

| Requirement       | Evidence / verification artifact              | Disposition                                              |
|-------------------|-----------------------------------------------|----------------------------------------------------------|
| FW-FR-001         | Figures 2-3; IPC source review                | Demonstrated                                             |
| FW-FR-002         | Shared-memory map; integrity counters         | Demonstrated with ring-capacity correction action        |
| FW-FR-003         | Figure 1 and Figure 4 GUI evidence            | Demonstrated                                             |
| FW-FR-004         | Figures 5-8 and scripted workflow             | Demonstrated                                             |
| FW-FR-005         | CSV/metadata workflow description             | Demonstrated                                             |
| FW-FR-006         | 25 kHz cap, 4 us high time, backlog bound     | Demonstrated by source review                            |
| FW-FR-007         | Heartbeat/fault/sequence/drop state           | Demonstrated; formal fault-injection witness recommended |
| FW-FR-008         | Stop/deadman behavior                         | Implemented; formal acceptance witness recommended       |
| FW-NFR-001 to 006 | Detailed design tables and result limitations | Mixed: demonstrated and controlled limitations           |

## Acceptance and Release Criteria

- A tagged target and host revision can execute the defined automated tests without an uncontrolled stop or pulse burst.
- Commanded and measured state remain visible, and the resulting CSV plus metadata can be traced to the run configuration.
- Heartbeat loss, sequence gaps, dropped records, and target faults are observable and produce the documented response.
- The shared-ring capacity correction is implemented or the reserved region is enlarged before controlled release.
- Claims remain limited to the demonstrated operating range and test evidence.

## 10. Limitations and Document Control

| Item               | Controlled statement                                                                                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prototype maturity | No formal shock, vibration, thermal, EMC, or long-duration certification is claimed.                                                                                |
| MotionIPC package  | Reviewed v0.1 sample producer is 100 Hz and should not be represented as the final integrated motor-control workload without the corresponding integrated revision. |
| Bandwidth claim    | Tested response is reported; formal bandwidth is not claimed.                                                                                                       |
| Release package    | Preserve tagged source, host build, test definitions, and representative datasets together.                                                                         |

### Attribution

Herder Elektronische Systemen developed the public-facing engineering package and documents the dual-core firmware integration, host software, test workflows, and measured evidence represented here. No third-party affiliation or endorsement is claimed.

### Revision record

| Revision | Date       | Description                                                                                                                                                                                                                                             |
|----------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.0      | 2026-08-05 | Reframed as a controlled FRD with explicit functional requirements, interfaces, acceptance criteria, traceability, while preserving GUI, architecture, and measured-result imagery; removed inherited layout defects and preserved evidence boundaries. |